

3.1 Introduction

The system control block includes several system features and control registers for a number of functions that are not related to specific peripheral devices. These include:

- Reset
- Brown-Out Detection
- External Interrupt Inputs
- Miscellaneous System Controls and Status

Each type of function has its own register(s) if any are required and unneeded bits are defined as reserved in order to allow future expansion. Unrelated functions never share the same register addresses

3.2 Pin description

[Table 6](#) shows pins that are associated with System Control block functions.

Table 6. Pin summary

Pin name	Pin direction	Pin description
EINT0	Input	External Interrupt Input 0 - An active low/high level or falling/rising edge general purpose interrupt input. This pin may be used to wake up the processor from Sleep, Deep-sleep, or Power-down modes.
EINT1	Input	External Interrupt Input 1 - See the EINT0 description above.
EINT2	Input	External Interrupt Input 2 - See the EINT0 description above.
EINT3	Input	External Interrupt Input 3 - See the EINT0 description above.
RESET	Input	External Reset input - A LOW on this pin resets the chip, causing I/O ports and peripherals to take on their default states, and the processor to begin execution at address 0x0000 0000.

3.3 Register description

All registers, regardless of size, are on word address boundaries. Details of the registers appear in the description of each function.

Table 7. Summary of system control registers

Name	Description	Access	Reset value	Address
External Interrupts				
EXTINT	External Interrupt Flag Register	R/W	0	0x400F C140
EXTMODE	External Interrupt Mode register	R/W	0	0x400F C148
EXTPOLAR	External Interrupt Polarity Register	R/W	0	0x400F C14C
Reset				
RSID	Reset Source Identification Register	R/W	see Table 8	0x400F C180
Syscon Miscellaneous Registers				
SCS	System Control and Status	R/W	0	0x400F C1A0

3.4 Reset

Reset has 4 sources on the LPC176x/5x: the $\overline{\text{RESET}}$ pin, Watchdog Reset, Power On Reset (POR), and Brown Out Detect (BOD).

The $\overline{\text{RESET}}$ pin is a Schmitt trigger input pin. Assertion of chip Reset by any source, once the operating voltage attains a usable level, starts the wake-up timer (see description in [Section 4.9 “Wake-up timer”](#) in this chapter), causing reset to remain asserted until the external Reset is de-asserted, the oscillator is running, a fixed number of clocks have passed, and the flash controller has completed its initialization. The reset logic is shown in the following block diagram (see [Figure 4](#)).

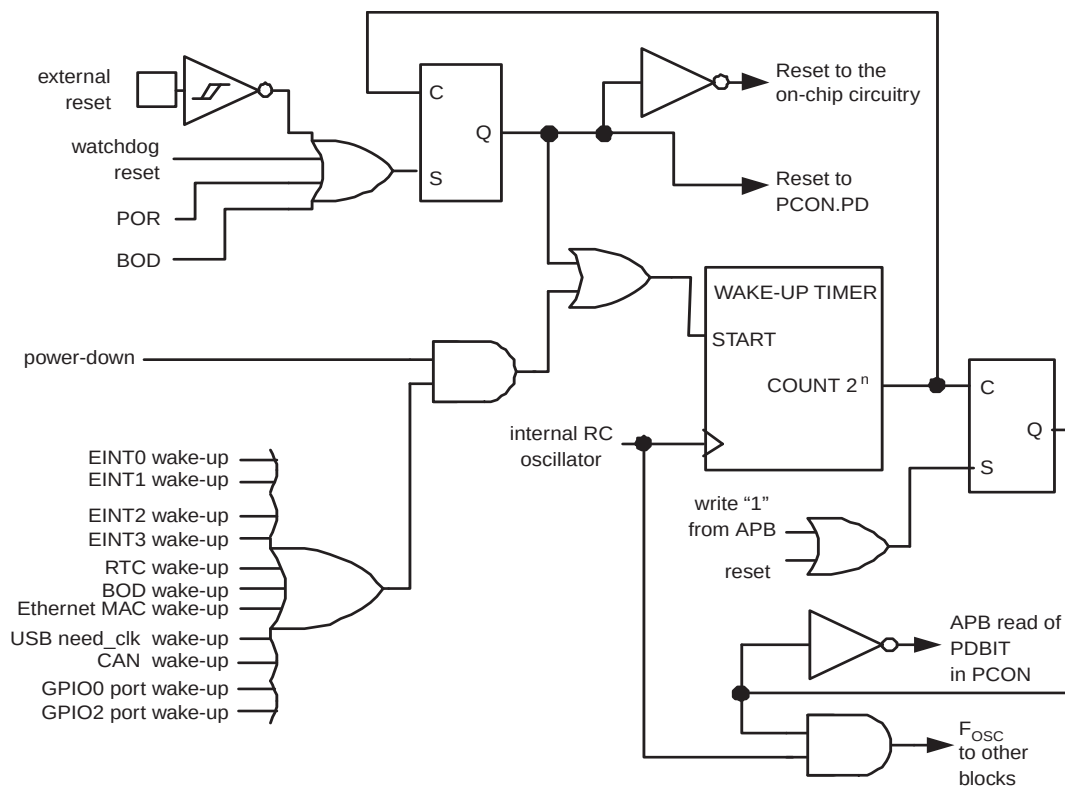


Fig 4. Reset block diagram including the wake-up timer

On the assertion of a reset source external to the Cortex-M3 CPU (POR, BOD reset, External reset, and Watchdog reset), the IRC starts up. After the IRC-start-up time (maximum of 60 μ s on power-up) and after the IRC provides a stable clock output, the reset signal is latched and synchronized on the IRC clock. Then the following two sequences start simultaneously:

1. The 2-bit IRC wake-up timer starts counting when the synchronized reset is de-asserted. The boot code in the ROM starts when the 2-bit IRC wake-up timer times out. The boot code performs the boot tasks and may jump to the flash. If the flash is not ready to access, the Flash Accelerator will insert wait cycles until the flash is ready.
2. The flash wake-up timer (9-bit) starts counting when the synchronized reset is de-asserted. The flash wake-up timer generates the 100 μ s flash start-up time. Once it times out, the flash initialization sequence is started, which takes about 250 cycles. When it's done, the Flash Accelerator will be granted access to the flash.

When the internal Reset is removed, the processor begins executing at address 0, which is initially the Reset vector mapped from the Boot Block. At that point, all of the processor and peripheral registers have been initialized to predetermined values.

[Figure 5](#) shows an example of the relationship between the $\overline{\text{RESET}}$, the IRC, and the processor status when the LPC176x/5x starts up after reset. See [Section 4.3.2 "Main oscillator"](#) for start-up of the main oscillator if selected by the user code.

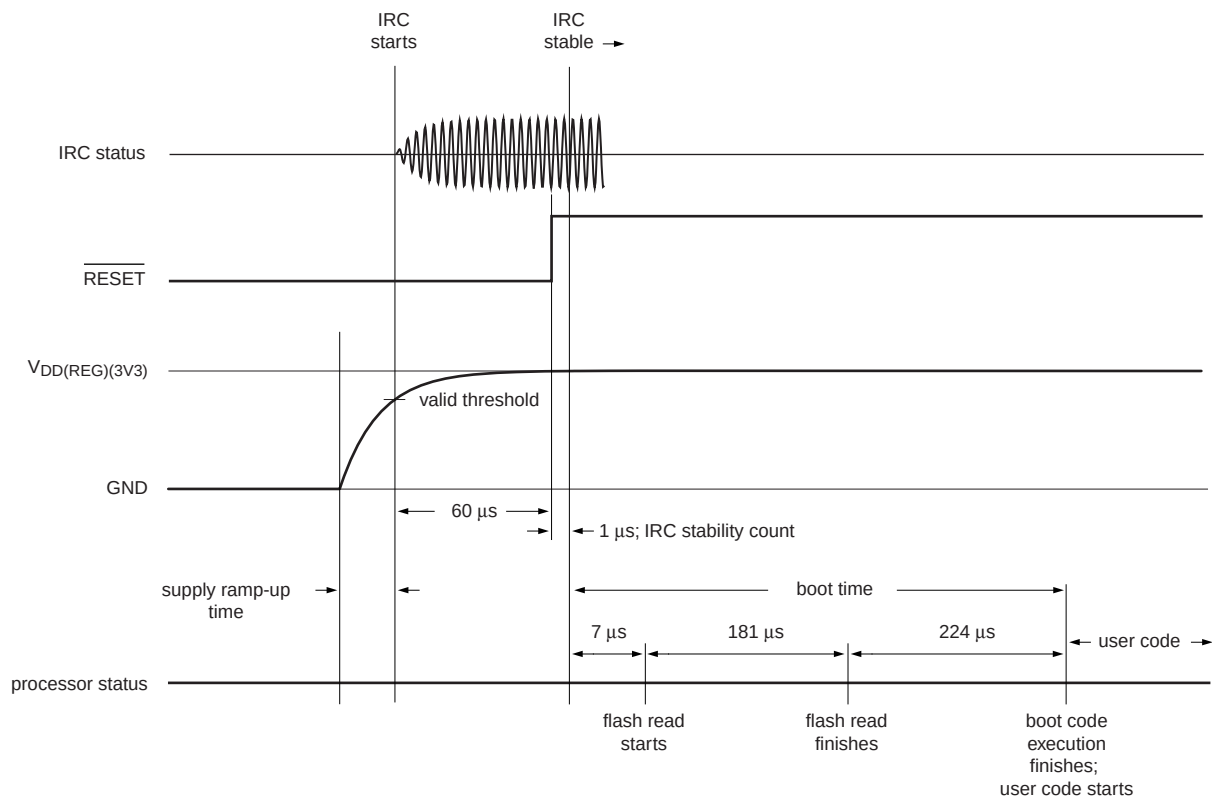


Fig 5. Example of start-up after reset

3.4.1 Reset Source Identification Register (RSID - 0x400F C180)

This register contains one bit for each source of Reset. Writing a 1 to any of these bits clears the corresponding read-side bit to 0. The interactions among the four sources are described below.

Table 8. Reset Source Identification register (RSID - address 0x400F C180) bit description

Bit	Symbol	Description	Reset value
0	POR	Assertion of the POR signal sets this bit, and clears all of the other bits in this register. But if another Reset signal (e.g., External Reset) remains asserted after the POR signal is negated, then its bit is set. This bit is not affected by any of the other sources of Reset.	See text
1	EXTR	Assertion of the $\overline{\text{RESET}}$ signal sets this bit. This bit is cleared only by software or POR.	See text
2	WDTR	This bit is set when the Watchdog Timer times out and the WDTRESET bit in the Watchdog Mode Register is 1. This bit is cleared only by software or POR. This bit is also cleared by an external reset.	See text
3	BODR	This bit is set when the $V_{\text{DD(REG)(3V3)}}$ voltage reaches a level below the BOD reset trip level (typically 1.85 V under nominal room temperature conditions). If the $V_{\text{DD(REG)(3V3)}}$ voltage dips from the normal operating range to below the BOD reset trip level and recovers, the BODR bit will be set to 1. If the $V_{\text{DD(REG)(3V3)}}$ voltage dips from the normal operating range to below the BOD reset trip level and continues to decline to the level at which POR is asserted (nominally 1 V), the BODR bit is cleared. If the $V_{\text{DD(REG)(3V3)}}$ voltage rises continuously from below 1 V to a level above the BOD reset trip level, the BODR will be set to 1. This bit is cleared only by software or POR. Note: Only in the case where a reset occurs and the POR = 0, the BODR bit indicates if the $V_{\text{DD(REG)(3V3)}}$ voltage was below the BOD reset trip level or not.	See text
4	SYSRESET	This bit is set if the processor has been reset due to a system reset request. Setting the SYSRESETREQ bit in the Cortex-M3 AIRCR register causes a chip reset in the LPC178x/177x. This bit is cleared only by software or POR.	See text
5	LOCKUP	This bit is set if the processor has been reset due to a "lockup". The lockup state causes a chip reset in the LPC178x/177x. This bit is cleared only by software or POR.	See text
31:6	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

3.5 Brown-out detection

The LPC176x/5x includes a Brown-Out Detector (BOD) that provides 2-stage monitoring of the voltage on the $V_{DD(REG)(3V3)}$ pins. If this voltage falls below the BOD interrupt trip level (typically 2.2 V under nominal room temperature conditions), the BOD asserts an interrupt signal to the NVIC. This signal can be enabled for interrupt in the Interrupt Enable Register in the NVIC in order to cause a CPU interrupt; if not, software can monitor the signal by reading the Raw Interrupt Status Register.

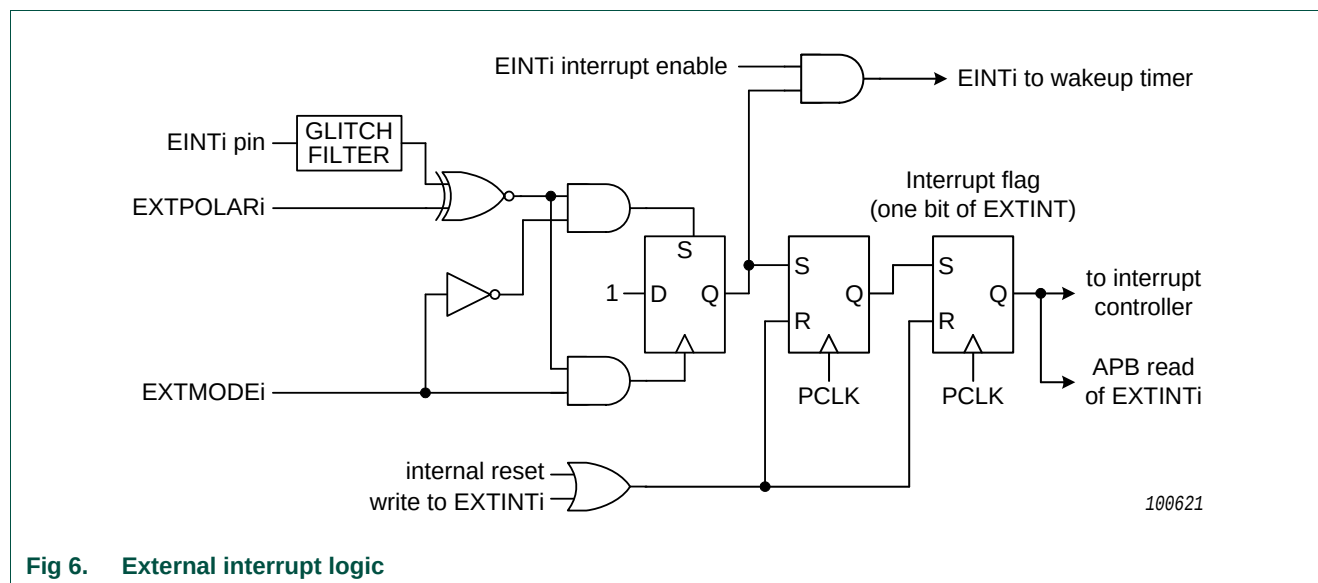
The second stage of low-voltage detection asserts Reset to inactivate the LPC176x/5x when the voltage on the $V_{DD(REG)(3V3)}$ pins falls below the BOD reset trip level (typically 1.85 V under nominal room temperature conditions). This Reset prevents alteration of the flash as operation of the various elements of the chip would otherwise become unreliable due to low voltage. The BOD circuit maintains this reset down below 1 V, at which point the Power-On Reset circuitry maintains the overall Reset.

Both the BOD reset interrupt level and the BOD reset trip level thresholds include some hysteresis. In normal operation, this hysteresis allows the BOD reset interrupt level detection to reliably interrupt, or a regularly-executed event loop to sense the condition.

But when Brown-Out Detection is enabled to bring the LPC176x/5x out of Power-down mode (which is itself not a guaranteed operation -- see [Section 4.8.7 "Power Mode Control register \(PCON - 0x400F C0C0\)"](#)), the supply voltage may recover from a transient before the wake-up timer has completed its delay. In this case, the net result of the transient BOD is that the part wakes up and continues operation after the instructions that set Power-down mode, without any interrupt occurring and with the BOD bit in the RSID being 0. Since all other wake-up conditions have latching flags (see [Section 3.6.2 "External Interrupt flag register \(EXTINT - 0x400F C140\)"](#) and [Section 27.6.2](#)), a wake-up of this type, without any apparent cause, can be assumed to be a Brown-Out that has gone away.

3.6 External interrupt inputs

The LPC176x/5x includes four External Interrupt Inputs as selectable pin functions. The logic of an individual external interrupt is represented in [Figure 6](#). In addition, external interrupts have the ability to wake up the CPU from Power-down mode. Refer to [Section 4.8.8 "Wake-up from Reduced Power Modes"](#) for details.



3.6.1 Register description

The external interrupt function has four registers associated with it. The EXTINT register contains the interrupt flags. The EXTMODE and EXTPOLAR registers specify the level and edge sensitivity parameters.

Table 9. External Interrupt registers

Name	Description	Access	Reset value ^[1]	Address
EXTINT	The External Interrupt Flag Register contains interrupt flags for EINT0, EINT1, EINT2 and EINT3. See Table 10 .	R/W	0x00	0x400F C140
EXTMODE	The External Interrupt Mode Register controls whether each pin is edge- or level-sensitive. See Table 11 .	R/W	0x00	0x400F C148
EXTPOLAR	The External Interrupt Polarity Register controls which level or edge on each pin will cause an interrupt. See Table 12 .	R/W	0x00	0x400F C14C

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

3.6.2 External Interrupt flag register (EXTINT - 0x400F C140)

When a pin is selected for its external interrupt function, the level or edge on that pin (selected by its bits in the EXTPOLAR and EXTMODE registers) will set its interrupt flag in this register. This asserts the corresponding interrupt request to the NVIC, which will cause an interrupt if interrupts from the pin are enabled.

Writing ones to bits EINT0 through EINT3 in EXTINT register clears the corresponding bits. In level-sensitive mode the interrupt is cleared only when the pin is in its inactive state.

Once a bit from EINT0 to EINT3 is set and an appropriate code starts to execute (handling wake-up and/or external interrupt), this bit in EXTINT register must be cleared. Otherwise event that was just triggered by activity on the EINT pin will not be recognized in future.

Important: whenever a change of external interrupt operating mode (i.e. active level/edge) is performed (including the initialization of an external interrupt), the corresponding bit in the EXTINT register must be cleared! For details see [Section 3.6.3 "External Interrupt Mode register \(EXTMODE - 0x400F C148\)"](#) and [Section 3.6.4 "External Interrupt Polarity register \(EXTPOLAR - 0x400F C14C\)"](#).

For example, if a system wakes up from Power-down using low level on external interrupt 0 pin, its post wake-up code must reset EINT0 bit in order to allow future entry into the Power-down mode. If EINT0 bit is left set to 1, subsequent attempt(s) to invoke Power-down mode will fail. The same goes for external interrupt handling.

More details on Power-down mode will be discussed in the following chapters.

Table 10. External Interrupt Flag register (EXTINT - address 0x400F C140) bit description

Bit	Symbol	Description	Reset value
0	EINT0	In level-sensitive mode, this bit is set if the EINT0 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT0 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state. [1]	0
1	EINT1	In level-sensitive mode, this bit is set if the EINT1 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT1 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state. [1]	0
2	EINT2	In level-sensitive mode, this bit is set if the EINT2 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT2 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state. [1]	0
3	EINT3	In level-sensitive mode, this bit is set if the EINT3 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT3 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state. [1]	0
31:4	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

[1] Example: e.g. if the EINTx is selected to be low level sensitive and low level is present on corresponding pin, this bit can not be cleared; this bit can be cleared only when signal on the pin becomes high.

3.6.3 External Interrupt Mode register (EXTMODE - 0x400F C148)

The bits in this register select whether each EINT pin is level- or edge-sensitive. Only pins that are selected for the EINT function (see [Section 8.5](#)) and enabled in the appropriate NVIC register) can cause interrupts from the External Interrupt function (though of course pins selected for other functions may cause interrupts from those functions).

Note: Software should only change a bit in this register when its interrupt is disabled in the NVIC (state readable in the ISERN/ICERN registers), and should write the corresponding 1 to EXTINT before enabling (initializing) or re-enabling the interrupt. An extraneous interrupt(s) could be set by changing the mode and not having the EXTINT cleared.

Table 11. External Interrupt Mode register (EXTMODE - address 0x400F C148) bit description

Bit	Symbol	Value	Description	Reset value
0	EXTMODE0	0	Level-sensitivity is selected for $\overline{\text{EINT0}}$.	0
		1	$\overline{\text{EINT0}}$ is edge sensitive.	
1	EXTMODE1	0	Level-sensitivity is selected for $\overline{\text{EINT1}}$.	0
		1	$\overline{\text{EINT1}}$ is edge sensitive.	
2	EXTMODE2	0	Level-sensitivity is selected for $\overline{\text{EINT2}}$.	0
		1	$\overline{\text{EINT2}}$ is edge sensitive.	
3	EXTMODE3	0	Level-sensitivity is selected for $\overline{\text{EINT3}}$.	0
		1	$\overline{\text{EINT3}}$ is edge sensitive.	
31:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

3.6.4 External Interrupt Polarity register (EXTPOLAR - 0x400F C14C)

In level-sensitive mode, the bits in this register select whether the corresponding pin is high- or low-active. In edge-sensitive mode, they select whether the pin is rising- or falling-edge sensitive. Only pins that are selected for the EINT function (see [Section 8.5](#)) and enabled in the appropriate NVIC register) can cause interrupts from the External Interrupt function (though of course pins selected for other functions may cause interrupts from those functions).

Note: Software should only change a bit in this register when its interrupt is disabled in the NVIC (state readable in the ISERN/ICERN registers), and should write the corresponding 1 to EXTINT before enabling (initializing) or re-enabling the interrupt. An extraneous interrupt(s) could be set by changing the polarity and not having the EXTINT cleared.

Table 12. External Interrupt Polarity register (EXTPOLAR - address 0x400F C14C) bit description

Bit	Symbol	Value	Description	Reset value
0	EXTPOLAR0	0	$\overline{\text{EINT0}}$ is low-active or falling-edge sensitive (depending on EXTMODE0).	0
		1	$\overline{\text{EINT0}}$ is high-active or rising-edge sensitive (depending on EXTMODE0).	
1	EXTPOLAR1	0	$\overline{\text{EINT1}}$ is low-active or falling-edge sensitive (depending on EXTMODE1).	0
		1	$\overline{\text{EINT1}}$ is high-active or rising-edge sensitive (depending on EXTMODE1).	
2	EXTPOLAR2	0	$\overline{\text{EINT2}}$ is low-active or falling-edge sensitive (depending on EXTMODE2).	0
		1	$\overline{\text{EINT2}}$ is high-active or rising-edge sensitive (depending on EXTMODE2).	

Table 12. External Interrupt Polarity register (EXTPOLAR - address 0x400F C14C) bit description

Bit	Symbol	Value	Description	Reset value
3	EXTPOLAR3	0	EINT3 is low-active or falling-edge sensitive (depending on EXTMODE3).	0
		1	EINT3 is high-active or rising-edge sensitive (depending on EXTMODE3).	
31:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

3.7 Other system controls and status flags

Some aspects of controlling LPC176x/5x operation that do not fit into peripheral or other registers are grouped here.

3.7.1 System Controls and Status register (SCS - 0x400F C1A0)

The SCS register contains several control/status bits related to the main oscillator. Since chip operation always begins using the Internal RC Oscillator, and the main oscillator may not be used at all in some applications, it will only be started by software request. This is accomplished by setting the OSCEN bit in the SCS register, as described in Table 3-13. The main oscillator provides a status flag (the OSCSTAT bit in the SCS register) so that software can determine when the oscillator is running and stable. At that point, software can control switching to the main oscillator as a clock source. Prior to starting the main oscillator, a frequency range must be selected by configuring the OSCRANGE bit in the SCS register.

Table 13. System Controls and Status register (SCS - address 0x400F C1A0) bit description

Bit	Symbol	Value	Description	Access	Reset value
3:0	-	-	Reserved. User software should not write ones to reserved bits. The value read from a reserved bit is not defined.	-	NA
4	OSCRANGE		Main oscillator range select.	R/W	0
		0	The frequency range of the main oscillator is 1 MHz to 20 MHz.		
		1	The frequency range of the main oscillator is 15 MHz to 25 MHz.		
5	OSCEN		Main oscillator enable.	R/W	0
		0	The main oscillator is disabled.		
		1	The main oscillator is enabled, and will start up if the correct external circuitry is connected to the XTAL1 and XTAL2 pins.		
6	OSCSTAT		Main oscillator status.	RO	0
		0	The main oscillator is not ready to be used as a clock source.		
		1	The main oscillator is ready to be used as a clock source. The main oscillator must be enabled via the OSCEN bit.		
31:7	-	-	Reserved. User software should not write ones to reserved bits. The value read from a reserved bit is not defined.	-	NA